

RELAYSPEC: ADAPTING FROZEN SPECULATIVE DRAFTERS WITH LINEAR MAPS

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ABSTRACT

As Large Language Models (LLMs) continue to advance, efficient inference remains a challenge. Speculative decoding reduces sequential generation by drafting several tokens for parallel target verification. Drafters such as EAGLE and DFlash use target hidden states, making reuse harder when the target changes. We introduce RelaySpec, which adapts a pretrained DFlash drafter by training five linear maps on its target-feature inputs while keeping the draft transformer frozen. For fine-tuned targets, we train the maps to improve token proposals using target-generated responses. For cross-model transfer, we train them to reconstruct the features expected by the drafter. The maps fold into the existing fusion projection before inference. We evaluate adaptation to fine-tuned targets, transfer across model sizes and families, and shared serving. A Qwen3-4B drafter transferred to Qwen3-8B reaches $5.82\times$ autoregressive throughput on MATH and $4.68\times$ on GSM8K, recovering 100.2% and 99.4% of native-drafter throughput. Transfer is weaker on code and chat. Comparisons with draft-body adaptation and feature interventions show when adapting the drafter’s inputs is useful and where its benefits are limited.

1 INTRODUCTION

As Large Language Models (LLMs) continue to advance, efficient inference remains a challenge. Autoregressive decoding generates one token at a time. This inherent seriality creates a major performance bottleneck: decoding is slow, memory-bound, and fails to fully utilize modern GPUs (Cai et al., 2024).

Speculative decoding accelerates a language model, called the *target*, using a smaller model or prediction module, called the *drafter*. The drafter proposes several candidate tokens, and the target checks them in parallel. Verification keeps an accepted prefix and replaces the first rejected proposal with a correction. Appropriate acceptance and correction rules preserve the target’s output distribution (Leviathan et al., 2023; Chen et al., 2023). Speedup depends on how many proposals are accepted and the time spent drafting and verifying them.

Some drafters also use the target’s hidden states, the internal vectors it computes while processing the preceding text. These vectors provide information about the context that can help the drafter propose tokens the target will accept. Medusa adds parallel prediction heads to the target’s final hidden states (Cai et al., 2024). The EAGLE family uses target features to guide autoregressive drafting (Li et al., 2024a;b; 2025). DFlash instead uses a block-diffusion drafter to predict several tokens in one pass, with target hidden states providing context to each draft layer (Chen et al., 2026).

This dependence complicates reuse when a service changes its target’s size or family, or fine-tunes it using methods such as LoRA (Hu et al., 2022). The new target can produce different hidden states and preferred continuations, and changed feature dimensions may be incompatible with the existing drafter. Training a replacement requires target-specific supervision, draft-weight optimization, and validation (Li et al., 2025; Chen et al., 2026). Repeating this process across target variants increases training and maintenance costs. Can we instead reuse an existing drafter by adapting only its target-feature inputs?

We introduce RelaySpec, a method for reusing a pretrained DFlash drafter after its target changes (Figure 1). The new target supplies hidden states from five layers, as defined in Section 2.1. A

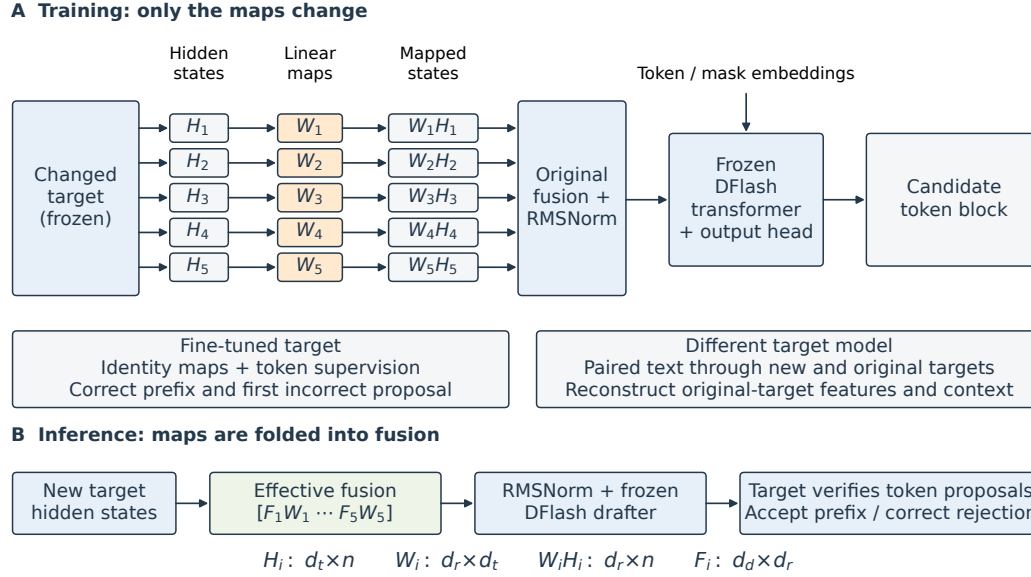


Figure 1: RelaySpec training and inference. Orange maps are trainable and blue model components are frozen. Each hidden-state matrix H_i enters a map W_i before fusion. A single position gives the matrix–vector product $W_i h_{i,t}$. Here n counts context positions, d_t is the new target’s feature width, d_r is the width expected by the drafter, and d_d is its hidden width. For Qwen3-8B with the Qwen3-4B drafter, $d_t = 4096$ and $d_r = d_d = 2560$. Each W_i is 2560×4096 , and exported fusion is 2560×20480 . The new target verifies proposals after adaptation.

separate linear map transforms each stream before the drafter’s original fusion projection combines them. The resulting context conditions the draft transformer, which predicts a block of candidate tokens. The new target verifies this block and accepts or corrects the proposals through the existing decoding procedure.

During adaptation, only the five maps are trained. The target, original fusion projection, draft transformer, normalization layers, and the embeddings and output head selected for the model pair remain frozen. A map can be rectangular when the target and drafter expect different feature dimensions. Before inference, we multiply each map into its corresponding fusion block, a composition of linear operations. This produces one effective projection, so inference needs no separate mapping operation.

For a LoRA-fine-tuned target, identity initialization leaves its hidden states unchanged and preserves the existing drafter’s behavior with that target. The maps therefore introduce no additional disruption at initialization. Training then modifies those hidden states so that they better condition the frozen drafter to predict the LoRA-adapted target’s tokens. We use *accept-until-fail (AUF)*: cross-entropy on saved target responses supervises the correct draft prefix and its first incorrect proposal (Yang & Li, 2026). Later positions are excluded because verification would stop at the earlier error.

For cross-model transfer, the drafter was trained to process hidden states from its original target. Replacing that target changes the features, and sometimes their dimensions, so the same drafter may no longer interpret them effectively. We run the original and new targets on the same text and train linear maps to reconstruct the original target’s features from the new target’s features. The *mean squared error (MSE)* objective has two terms: one compares each mapped feature with its original-target counterpart, and the other compares their combined context after the frozen fusion projection and RMSNorm. The context term supervises what the draft layers actually receive, beyond matching each layer separately. Both terms use squared errors scaled by the corresponding original-target vector’s squared norm, with numerical stabilization.

We evaluate specialization, cross-model transfer, training budgets, and shared serving. Matched comparisons include draft-body updates, direct fusion updates, and different token objectives. Their

results delimit the method: body adaptation or decaying cross-entropy can be stronger on some workloads, and a pooled adapter can approach separate task adapters. We therefore measure both single-request and shared-serving performance rather than assume that every target needs a separate drafter or that higher acceptance always gives higher throughput.

We analyze the learned maps to understand why adaptation helps. First, we use singular value decomposition (SVD) to retain only the largest components of either the full map W_i or its correction $\Delta_i = W_i - I$. For the tested GSM8K and NanoCoder checkpoints, truncating the full map to rank one or two sharply reduces acceptance. Using $I + [\Delta_i]_r$, where $[\Delta_i]_r$ is the rank- r truncated correction, preserves more acceptance, though these low ranks do not recover the full-map result. Second, scaling only the correction shows that half strength retains much of the benefit, while stronger corrections reduce acceptance. Third, we compare the correction with the LoRA-induced change in fused hidden states. Their mean cosine similarities are close to zero and similar to shuffled-position controls, providing no evidence of systematic cancellation of the LoRA change. Finally, in cross-size transfer, moving token-trained maps halfway toward an MSE-trained solution improves acceptance and throughput on the tested Math and Code workloads. This supports learning inputs the frozen drafter can use, without establishing a universal advantage for MSE.

Our contributions are:

- We propose RelaySpec, which adapts a pretrained DFlash drafter through five linear maps while freezing its transformer. The maps fold into the original fusion projection, and we study token supervision for target specialization and feature reconstruction for cross-model reuse.
- We evaluate adaptation across fine-tuned targets, model sizes, and families, including training-budget and shared-serving comparisons. On Qwen3-4B LoRA targets, the selected AUF maps improve mean per-request throughput over the unchanged DFlash drafter by 37.58% and 89.52% on GSM8K and KiCad, respectively. The selected Qwen3-4B-to-8B transfer recovers 100.2% and 99.4% of native-drafter throughput on MATH and GSM8K, with weaker transfer on code and chat.
- We analyze full-map and correction-map SVD truncation, correction scaling, alignment with LoRA-induced feature changes, and interpolation toward MSE-trained maps.

2 PRELIMINARIES

2.1 TARGETS, DRAFTERS, AND NOTATION

The *original target* is the model used to train the available drafter. The *new target* is the model to be accelerated. A *native drafter* is trained for the target being evaluated. In cross-model transfer, the original target used for feature reconstruction and the native drafter used as a performance baseline are therefore different objects.

DFlash predicts a block of candidate tokens from an anchor token and masked future positions in one forward pass. It fuses selected target hidden states and uses the resulting context to supply keys and values to every draft layer (Chen et al., 2026). Our pretrained checkpoints use five feature streams. We write the original target’s feature at stream i and position t as $h_{i,t}^{(o)} \in \mathbb{R}^{d_r}$. The original fusion blocks $F_i \in \mathbb{R}^{d_d \times d_r}$ produce

$$c_t^{(r)} = N \left(\sum_{i=1}^5 F_i h_{i,t}^{(o)} \right), \quad F = [F_1 \cdots F_5]. \quad (1)$$

Here d_d is the draft hidden width and N is the original RMSNorm, which normalizes vector magnitude and applies learned coordinate-wise gains (Zhang & Sennrich, 2019). For the new target, $h_{i,t} \in \mathbb{R}^{d_t}$ and $H_i = [h_{i,1} \cdots h_{i,n}] \in \mathbb{R}^{d_t \times n}$. We use column vectors throughout. A RelaySpec map has shape $W_i \in \mathbb{R}^{d_r \times d_t}$, so $W_i h_{i,t}$ can enter the original fusion block. The widths d_t , d_r , and d_d need not be equal.

2.2 VERIFICATION AND USEFUL TRAINING POSITIONS

Let $x_{1:t}$ be an accepted prefix. The drafter proposes K tokens $y_{1:K}$ and the target scores them in parallel under $x_{1:t}, y_{<j}$. With draft distribution q and target distribution p over a shared vocabulary, speculative sampling accepts y_j with probability

$$a_j = \min\left(1, \frac{p(y_j \mid x_{1:t}, y_{<j})}{q(y_j \mid x_{1:t}, y_{<j})}\right). \quad (2)$$

At rejection, a correction is sampled from the normalized positive part of $p - q$. If all proposals are accepted, the target provides a further token. Using the actual proposal probabilities preserves the target distribution in exact arithmetic (Leviathan et al., 2023; Chen et al., 2023). Our experiments use greedy decoding: retain the longest prefix matching the target’s greedy choices, then use the target’s correction.

Only a consecutive prefix contributes accepted draft tokens. If $L \in \{0, \dots, K\}$ is its length, then $\mathbb{E}[L] = \sum_{j=1}^K \Pr(L \geq j)$. This motivates supervising the first incorrect proposal as well as the correct prefix. AUF applies that rule to comparisons against saved target tokens (Yang & Li, 2026). Its training-time correct prefix is a prediction diagnostic, not a direct measurement of live decoding speed.

2.3 ACCESS, ALIGNMENT, AND MEASUREMENT

Both settings require access to the new target’s hidden states. Specialization starts from the existing same-model drafter with identity maps. Cross-model reconstruction additionally runs the original target on the same text, using aligned positions as supervision. The original target is needed for this data preparation, not adapted inference. Only features available from the current prefix may condition a draft block. The target remains frozen during drafter adaptation, including any active task adapter.

Shared token IDs allow direct verification. Different tokenizers require a separate conversion and alignment procedure, which the hidden-state maps do not supply. Our cross-tokenizer evaluation includes this bridge’s execution cost. Greedy bridge results do not establish stochastic distribution preservation, and exact-arithmetic guarantees do not imply bitwise agreement across numerical kernels.

We distinguish accepted draft tokens from emitted tokens, which include target corrections or bonus tokens. We also distinguish mean per-request token rates, total tokens divided by summed request time, and aggregate tokens divided by workload wall time under concurrency. These measurements have different denominators and are compared only under matched conditions. Reconstruction error and offline correct-prefix length diagnose the maps, while live acceptance and timing establish decoding performance.

3 RELATED WORK

Target-conditioned drafting. Medusa predicts future positions with additional heads on the target’s final hidden states (Cai et al., 2024). EAGLE uses target features for autoregressive drafting, with later versions improving tree construction and training (Li et al., 2024a;b; 2025). ReDrafter combines target-conditioned recurrent drafting with beam search and dynamic tree attention (Cheng et al., 2024). DFlash predicts a masked block in parallel and supplies target context to every draft layer (Chen et al., 2026). RelaySpec starts from a pretrained DFlash checkpoint and studies its reuse after the target changes.

Adapting to changed targets. DistillSpec studies training data and divergence objectives for matching a drafter to its target (Zhou et al., 2024). Online Speculative Decoding updates the drafter from serving feedback (Liu et al., 2024). Efficient Draft Adaptation separates shared and target-specific components and trains the private component using regenerated and selected data (Lin et al., 2026). OmniDraft addresses vocabulary mismatch with an online cache and hybrid distillation (Ramakrishnan et al., 2025). These methods already address parts of the reuse problem. RelaySpec

asks how much adaptation can be placed in five maps before a pretrained DFlash fusion projection while retaining the draft transformer.

Parallel drafting across targets. PARD trains an autoregressive model for parallel prediction and supports reuse within a target family (An et al., 2026a). PARD-2 supports both target-conditioned and target-independent drafting (An et al., 2026b). RelaySpec instead starts with a target-conditioned DFlash checkpoint and adapts it to changed targets. Our experiments compare separate and pooled adaptation because target-specific training is not automatically necessary or better.

Steering pretrained drafters. SD² uses linear projections of target hidden states to add steering signals inside a pretrained autoregressive drafter’s feed-forward layers (Berdoz et al., 2026). Its main method trains the steering projections and draft weights jointly. Its frozen-drafter ablation also improves acceptance, although joint training performs better in that comparison. This establishes prior evidence for steering both trainable and frozen drafters. RelaySpec studies DFlash input maps that fold into the fusion projection, comparing token supervision for specialization with reconstruction for cross-model transfer.

Linear maps and training-time parameterization. Huang et al. (2025) use linear transformations to transfer concept-steering vectors across language models. Chen et al. (2025) learn affine maps between model activations to transfer features, predictors, and steering vectors. These results motivate mapping hidden states, but do not establish that a mapped drafter retains its acceptance or speed. RepSpec adds redundant linear structures during draft training and merges them for inference (Huo et al., 2026). RelaySpec shares the use of mergeable linear operations, but trains maps on target-feature inputs while preserving the pretrained draft transformer. Our direct-fusion comparisons and matched-weight gradient diagnostics examine how this parameterization changes optimization. They do not attribute a trained checkpoint’s advantage to a single gradient measurement.

4 METHOD

5 EXPERIMENTS

6 ANALYSIS

7 CONCLUSION

AI USE STATEMENT

ETHICS STATEMENT

REPRODUCIBILITY STATEMENT

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